الأسئلة المقالية — 10 متقدم · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (54 سؤالاً)

LO21 التحليل باستعمال المتطابقات الخاصة · Special-product factorization

21.1 LO21 · Reveal p. 680 · 35

Factor completely using integers.

$$-72p^2 + 8x^2$$

a. Show your factorisation and check by expansion. (10)

الأستاذ كامل بشارة 0502837825

21.2 LO21 · Reveal p. 680 · 38

Factor completely using integers.

$$16b^2 - 100$$

a. Show your factorisation and check by expansion. (10)

الأستاذ كامل بشارة 0502837825

21.3 LO21 · Reveal p. 681 · 51

Determine whether this is a perfect-square trinomial; if so, factor it.

$$9z^2 - 6z + 1$$

a. Show your factorisation and check by expansion. (10)

21.4 LO21 · Reveal p. 681 · 52

Determine whether this is a perfect-square trinomial; if so, factor it.

$$36x^2 - 60x + 25$$

a. Show your factorisation and check by expansion. (10)

21.5 LO21 · Reveal p. 680 · 36

Factor completely using integers.

$$20q^2 - 5r^2$$

a. Show your factorisation and check by expansion. (10)

21.6 LO21 · Reveal p. 680 · 37

Factor completely using integers.

$$32a^2 - 50b^2$$

a. Show your factorisation and check by expansion. (10)

21.7 LO21 · Reveal p. 680 · 39

Factor completely using integers.

$$49x^2 - 64y^2$$

a. Show your factorisation and check by expansion. (10)

21.8 LO21 · Reveal p. 680 · 40

Factor completely using integers.

$$3n^4 - 42n^3 + 147n^2$$

a. Show your factorisation and check by expansion. (10)

21.9 LO21 · Reveal p. 680 · 41

Factor completely using integers.

$$8m^3 - 24m^2 + 18m$$

a. Show your factorisation and check by expansion. (10)

21.10 LO21 · Reveal p. 681 · 53

Determine whether this is a perfect-square trinomial; if so, factor it.

$$49r^2 + 14r + 4$$

a. Show your factorisation and check by expansion. (10)

21.11 LO21 · Reveal p. 681 · 54

Determine whether this is a perfect-square trinomial; if so, factor it.

$$a^2 + 14a + 49$$

a. Show your factorisation and check by expansion. (10)

21.12 LO21 · Reveal p. 681 · 55

Determine whether this is a perfect-square trinomial; if so, factor it.

$$t^2 - 18t + 81$$

a. Show your factorisation and check by expansion. (10)

21.13 LO21 · Reveal p. 681 · 56

Determine whether this is a perfect-square trinomial; if so, factor it.

$$4c^2 + 2cd + d^2$$

a. Show your factorisation and check by expansion. (10)

LO22 الاستدلال بالزوايا الداخلية والخارجية · Interior and exterior angle reasoning

22.1 LO22 · Reveal p. 64 · 9

The given angle is an interior angle of a regular polygon. Find the number of sides.

$$144^\circ$$

a. Show an equation and solve for n. (10)

22.2 LO22 · Reveal p. 64 · 11

The given angle is an interior angle of a regular polygon. Find the number of sides.

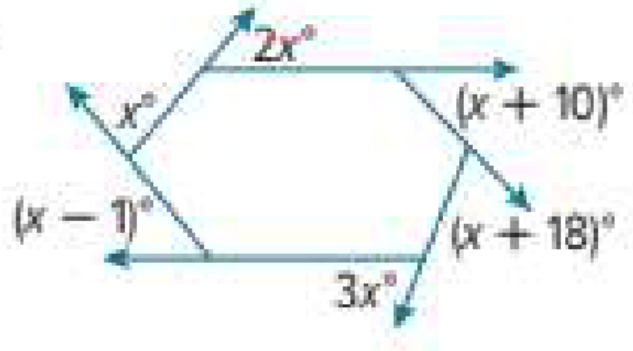
$160°$

a. Show an equation and solve for n. (10)

22.3 LO22 · Reveal p. 64 · 15

Find x in the diagram.

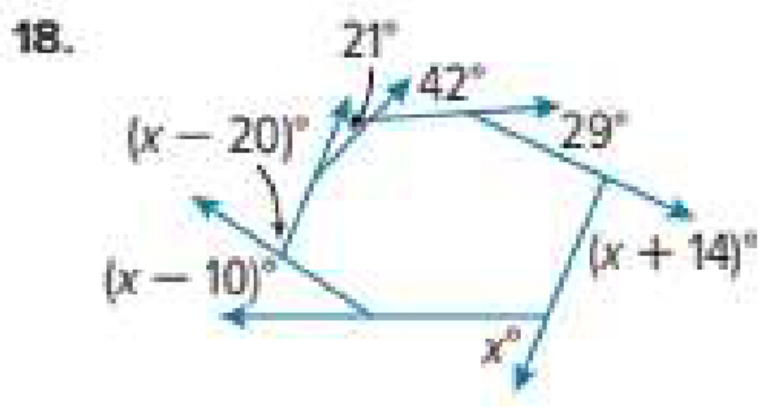
15.



a. Show how you use the exterior-angle sum. (10)

22.4 LO22 · Reveal p. 64 · 18

Find x in the diagram.



a. Show how you use the exterior-angle sum. (10)

22.5 LO22 · Reveal p. 64 · 10

The given angle is an interior angle of a regular polygon. Find the number of sides.

$156°$

a. Show an equation and solve for n. (10)

22.6 LO22 · Reveal p. 64 · 12

The given angle is an interior angle of a regular polygon. Find the number of sides.

$108°$

a. Show an equation and solve for n. (10)

22.7 LO22 · Reveal p. 64 · 13

The given angle is an interior angle of a regular polygon. Find the number of sides.

$120°$

a. Show an equation and solve for n. (10)

22.8 LO22 · Reveal p. 64 · 14

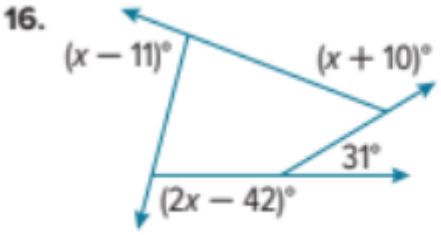
The given angle is an interior angle of a regular polygon. Find the number of sides.

$150°$

a. Show an equation and solve for n. (10)

22.9 LO22 · Reveal p. 64 · 16

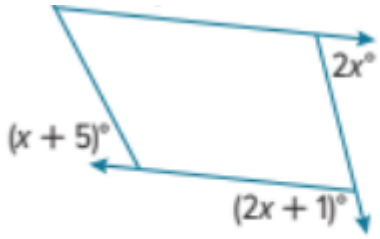
Find x in the diagram.



a. Show how you use the exterior-angle sum. (10)

22.10 LO22 · Reveal p. 64 · 17

Find x in the diagram.

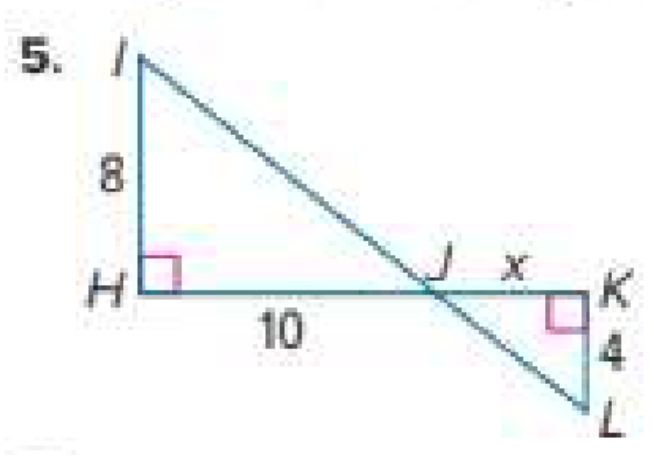


a. Show how you use the exterior-angle sum. (10)

LO23 حل المسائل باستعمال المثلثات المتشابهة · Solving with similar triangles

23.1 LO23 · Reveal p. 139 · 5

Identify the similar triangles. Then find x.

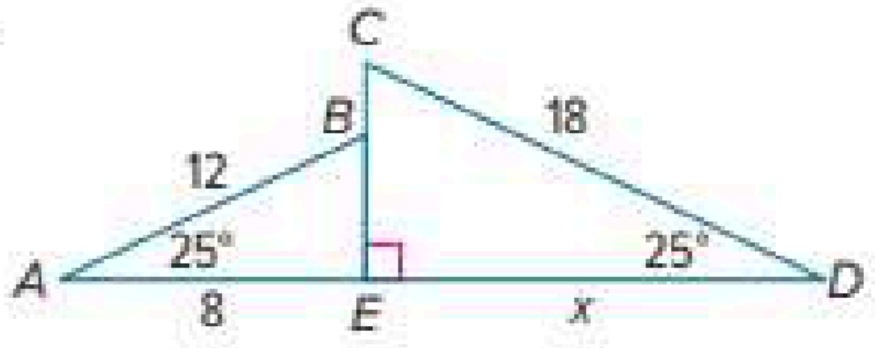


a. State corresponding vertices and show a proportion. (10)

23.2 LO23 · Reveal p. 139 · 6

Identify the similar triangles. Then find x.

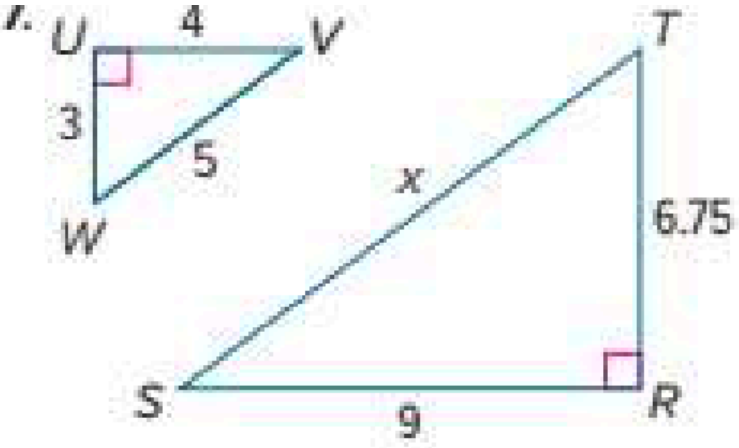
6.



a. State corresponding vertices and show a proportion. (10)

23.3 LO23 · Reveal p. 139 · 7

Identify the similar triangles. Then find x.

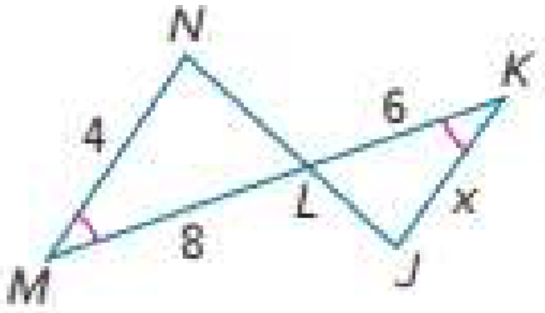


a. State corresponding vertices and show a proportion. (10)

23.4 LO23 · Reveal p. 139 · 8

Identify the similar triangles. Then find x.

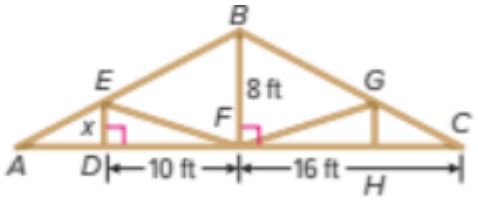
3.



a. State corresponding vertices and show a proportion. (10)

23.5 LO23 · Reveal p. 139 · 9

The skeleton of a roof is shown. Find x such that triangles DEF and FBC in the outline of the roof are similar.



a. Write the proportion and solve. (10)

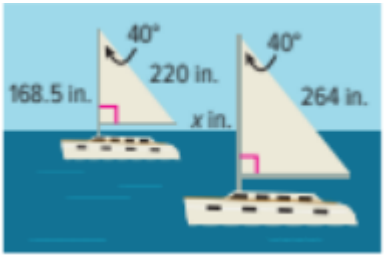
23.6 LO23 · Reveal p. 139 · 10

A radio tower casts an 8-foot-long shadow at the same time that a vertical yardstick casts a shadow one half inch long. If the triangles formed by the objects and their shadows are similar, how tall is the radio tower?

a. Write the proportion and solve. (10)

23.7 LO23 · Reveal p. 139 · 11

The two sailboats shown are participating in a regatta. If the sails are similar, what is x?



a. Write the proportion and solve. (10)

LO24 المسافة ومنتصف القطعة في الفراغ · Distance and midpoint in space

24.1 LO24 · Reveal p. 177 · 13

Find the midpoint of the segment joining the two points.

$$K(-2,-4,-4), \quad L(4,2,0)$$

a. Determine the coordinates of midpoint M. (10)

24.2 LO24 · Reveal p. 177 · 15

Find the midpoint of the segment joining the two points.

$$R(3, 3, 4), \quad V(5, 4, 13)$$

a. Determine the coordinates of midpoint M. (10)

24.3 LO24 · Reveal p. 178 · 19

Find the distance and midpoint of the segment.

$$P(-5, -2, -1), \quad Q(-1, 0, 3)$$

a. Determine the distance. (5)

b. Determine the coordinates of midpoint M. (5)

24.4 LO24 · Reveal p. 178 · 23

Find the distance and midpoint of the segment.

$$B(\sqrt{3}, 2, 2\sqrt{2}), \quad C(-2\sqrt{3}, 4, 4\sqrt{2})$$

a. Determine the distance. (5)

b. Determine the coordinates of midpoint M. (5)

24.5 LO24 · Reveal p. 177 · 14

Find the midpoint of the segment joining the two points.

$$W(-1, -3, -6), \quad Z(-1, 5, 10)$$

a. Determine the coordinates of midpoint M. (10)

24.6 LO24 · Reveal p. 177 · 16

Find the midpoint of the segment joining the two points.

$$A(4, 6, -8), \quad B(0, 0, 0)$$

a. Determine the coordinates of midpoint M. (10)

24.7 LO24 · Reveal p. 177 · 17

Find the midpoint of the segment joining the two points.

$$C(8, 7, 11), \quad D(2, 1, 8)$$

a. Determine the coordinates of midpoint M. (10)

24.8 LO24 · Reveal p. 177 · 18

Find the midpoint of the segment joining the two points.

$$T(-1, -7, 9), \quad U(5, -1, -6)$$

a. Determine the coordinates of midpoint M. (10)

24.9 LO24 · Reveal p. 178 · 20

Find the distance and midpoint of the segment.

$$J(1, 1, 1), \quad K(-1, -1, -1)$$

a. Determine the distance. (5)

b. Determine the coordinates of midpoint M. (5)

24.10 LO24 · Reveal p. 178 · 21

Find the distance and midpoint of the segment.

$F(\frac{3}{5}, 0, \frac{4}{5}), \quad G(0, 3, 0)$

a. Determine the distance. (5)

b. Determine the coordinates of midpoint M. (5)

24.11 LO24 · Reveal p. 178 · 22

Find the distance and midpoint of the segment.

$G(1, -1, 6), \quad H(\frac{1}{5}, -\frac{2}{5}, 2)$

a. Determine the distance. (5)

b. Determine the coordinates of midpoint M. (5)

24.12 LO24 · Reveal p. 178 · 24

Find the distance and midpoint of the segment.

$S(6\sqrt{3}, 4, 4\sqrt{2}), \quad T(4\sqrt{3}, 5, \sqrt{2})$

a. Determine the distance. (5)

b. Determine the coordinates of midpoint M. (5)

25.1 LO25 · Reveal p. 208 · 15

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 50^\circ, \quad a = 34, \quad b = 40$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.2 LO25 · Reveal p. 208 · 16

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 24^\circ, \quad a = 3, \quad b = 8$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.3 LO25 · Reveal p. 208 · 17

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 125^\circ, \quad a = 22, \quad b = 15$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.4 LO25 · Reveal p. 208 · 19

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 30^\circ, \quad a = 2, \quad b = 4$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.5 LO25 · Reveal p. 208 · 18

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 30^\circ, \quad a = 1, \quad b = 4$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.6 LO25 · Reveal p. 208 · 20

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 30^\circ, \quad a = 3, \quad b = 4$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.7 LO25 · Reveal p. 208 · 21

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 38^\circ, \quad a = 10, \quad b = 9$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.8 LO25 · Reveal p. 208 · 22

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 78^\circ, \quad a = 8, \quad b = 5$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.9 LO25 · Reveal p. 208 · 23

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 133^\circ, \quad a = 9, \quad b = 7$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.10 LO25 · Reveal p. 208 · 24

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 127^\circ, \quad a = 2, \quad b = 6$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.11 LO25 · Reveal p. 208 · 25

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 109^\circ, \quad a = 24, \quad b = 13$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)

25.12 LO25 · Reveal p. 208 · 26

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 48^\circ, \quad a = 11, \quad b = 16$$

a. Determine the number of solutions and justify it. (4)

b. Give every possible triangle, or explain why none exists. (6)